UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A2	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A3	FALSE	3D animation does not use modelling.	3D animation relies on models.
A4	FALSE	Anti-aliasing makes the edges of text look more jagged.	Anti-aliasing smooths jagged edges.
A5	TRUE	A storyboard is a visual plan of a multimedia sequence.	Storyboards show the planned flow.
A6	TRUE	A higher sampling rate generally produces a more accurate recording.	More samples capture more detail.
A7	TRUE	Cel animation is created by drawing each frame by hand.	Cel animation uses hand-drawn frames.
A8	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A9	FALSE	Higher resolution always means a smaller file.	Higher resolution usually increases file size.
A10	TRUE	USB is a plug-and-play connection standard.	USB devices are recognised automatically.
A11	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A12	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A13	TRUE	Path animation moves an object along a predefined route.	Path animation follows a motion path.
A14	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A15	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	H	Serif	font with small strokes at character ends
B2	F	Scanner	input device that digitises documents
B3	N	Sampling	converting sound into digital values
B4	I	Browser	software that displays web pages
B5	G	Multimedia	integration of text, graphics, audio, video and animation
B6	L	HTML	language for web structure
B7	A	Frame rate	frames displayed per second
B8	M	Linear multimedia	presentation without user control of sequence
B9	C	Chroma key	replacing a colour background
B10	O	Hypermedia	linked multimedia elements
B11	D	MIDI	standard for musical notes and instructions
B12	J	Streaming	playing media while it downloads
B13	E	Amplitude	height of a sound wave
B14	K	Storyboard	visual sequence plan
B15	B	Sans serif	font without small strokes

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	D	Vector
C2	B	Non-linear
C3	C	Smooth jagged edges
C4	B	A cinema presentation
C5	B	CMYK
C6	A	Along a predefined route
C7	B	Number of pixels in an image
C8	A	Display web pages
C9	A	Messages with a beginning and end
C10	C	Play while downloading
C11	A	Plotter
C12	D	Structure web content
C13	D	Analog sound to digital values
C14	C	Pixels
C15	C	Temporary working memory
C16	A	Frames per second
C17	D	An early model of a product
C18	A	Automatic device recognition
C19	A	Permission to use a work
C20	A	A simulated environment
C21	B	Important frames in an animation
C22	D	Merging of media and computing
C23	C	In-between frames
C24	D	Multimedia with hyperlinks
C25	A	Drawing each frame by hand
C26	A	A low-fidelity layout sketch
C27	A	Improves accuracy
C28	B	Scanner
C29	A	Printed output
C30	A	Card-based

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PART D - WORKED ANSWERS

QUESTION 1

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(5 marks)

QUESTION 2

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 3

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(4 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

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QUESTION 4

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(3 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

END OF MARKING SCHEME

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